

> Hardening Email Sender Authentication:
A Lab-Based Evaluation of SPF/DKIM/DMARC
Misconfigurations

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> Introduction

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Why does Email Sender Authentication Matter?

- Email remains a primary vector for phishing attacks and weak sender-authentication deployment can create exploits for attackers to appear trustworthy.
- SPF, DKIM, and DMARC were designed to mitigate spoofing and throughout this project we study how their misconfiguration can create vulnerabilities.
- We also aim to propose ways to mitigate the impact of the possible attacks that could stem from these vulnerabilities.

> Objectives

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> Objectives

What we want to show

- Reproduce sender spoofing in a safe, isolated environment that isolates the attack surface from the real internet.
- Demonstrate a working SPF-based spoofing path using the attacker container against the victim mail server.
- Document the setup clearly so the lab can be extended later for DKIM, DMARC, and hardening tests.

Expected outcome

- A reproducible lab that shows how SPF misconfiguration enables spoofing and how to block it.
- Concrete DNS and MTA hardening steps that can be validated in the same environment.

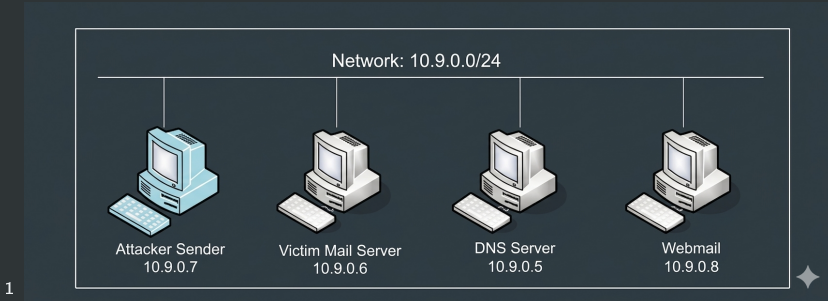
> Preliminary solution concept

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> Preliminary solution concept

Lab design

- The lab uses a DNS server, a victim mail server, and an attacker container.
- DNS is configured so SPF-related misconfiguration allows spoofed mail to pass.
- The current stage focuses on the SPF scenario that already succeeds; DKIM and DMARC remain for later validation.



¹Image generated by Google's Gemini Pro Model, 2026. [1]

> Development methodology and key tools

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> Development methodology and key tools

Implementation approach

- Docker Compose is used to orchestrate the lab environment.
- BIND provides the DNS layer, docker-mailserver hosts the victim mail system, and Espoofer runs from the attacker container.
- Logs and message delivery behavior are used to check whether the spoofed SPF case succeeds.

```
version: "3"
services:
  dns-server:      # BIND
  victim-mail:     # Postfix/Dovecot
  attacker-sender: # Espoofer + custom scripts
  webmail:         # User interface for the email service
```

> Next steps and proposed timeline

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> Next steps and proposed timeline

Short-term plan

- Stabilize and document the SPF spoofing scenario.
- Test hardening measures at the DNS and MTA levels.
- Revisit DKIM and DMARC later as future extensions.

Proposed timeline

- Short term: validate the setup and confirm SPF results.
- Mid term: compare mitigation strategies and collect evidence.
- Final phase: extend the lab and summarize recommendations.

> References

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